

VISION Resistivity
Dual Frequency
Recorded Mode

Schlumberger

Country: Norway

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

UWID:

Longitude: 3° 26' 36.13" E

Rig Name:

West Hercules

Block: 31/5

Rig Type:

Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 12 1/4 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level

Ground Level: 307.00 m

Acquisition Dates: 23-Dec-2019 -- 25-Dec-2019

Other Services:

Log Interval: 1856.00(m) -- 2584.00(m)

Direction & Inclination (Surveys)

Index Types: Measured Depth

Directional Drilling

Index Scales: 1:500

Annular Pressure While Drilling

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

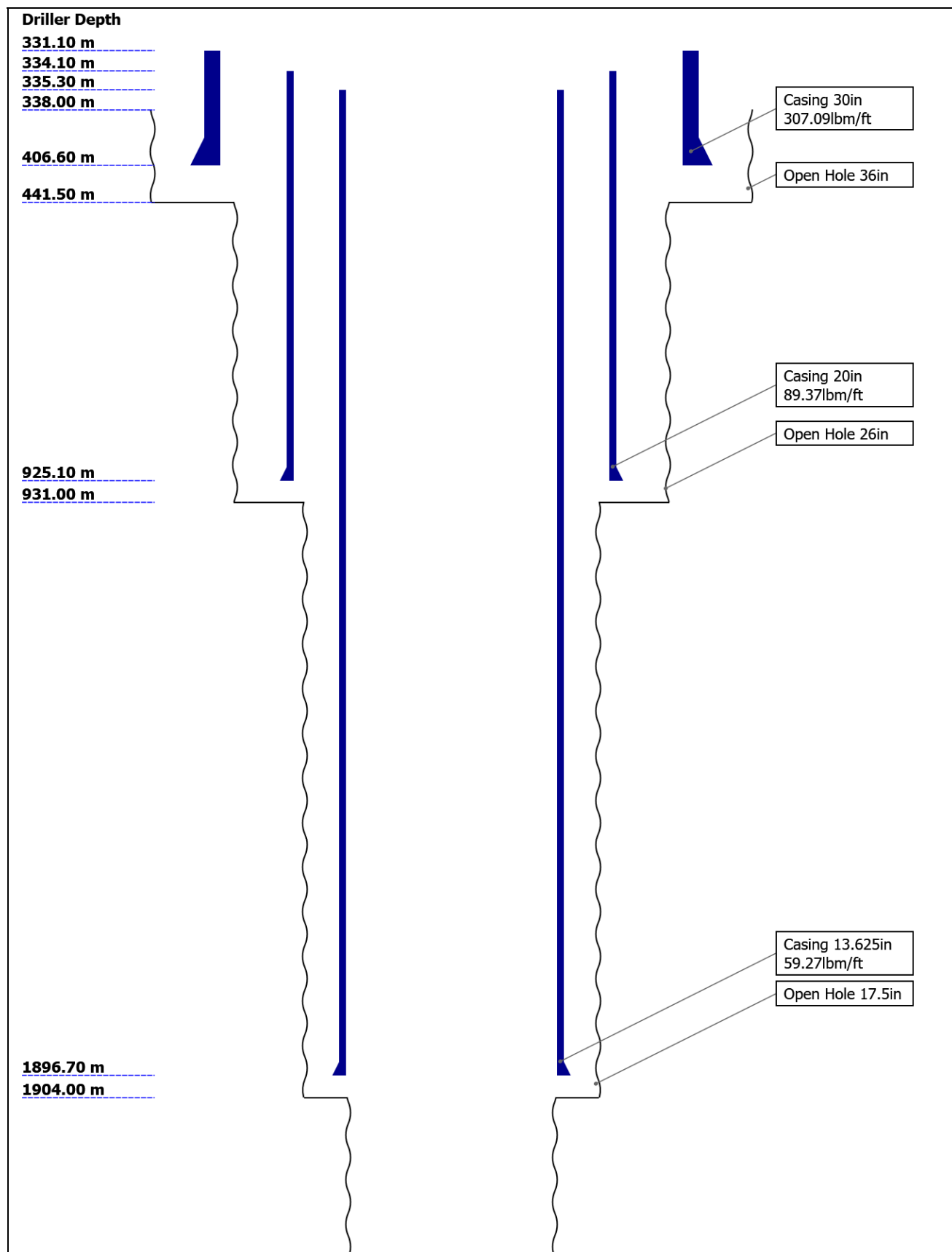
Disclaimer

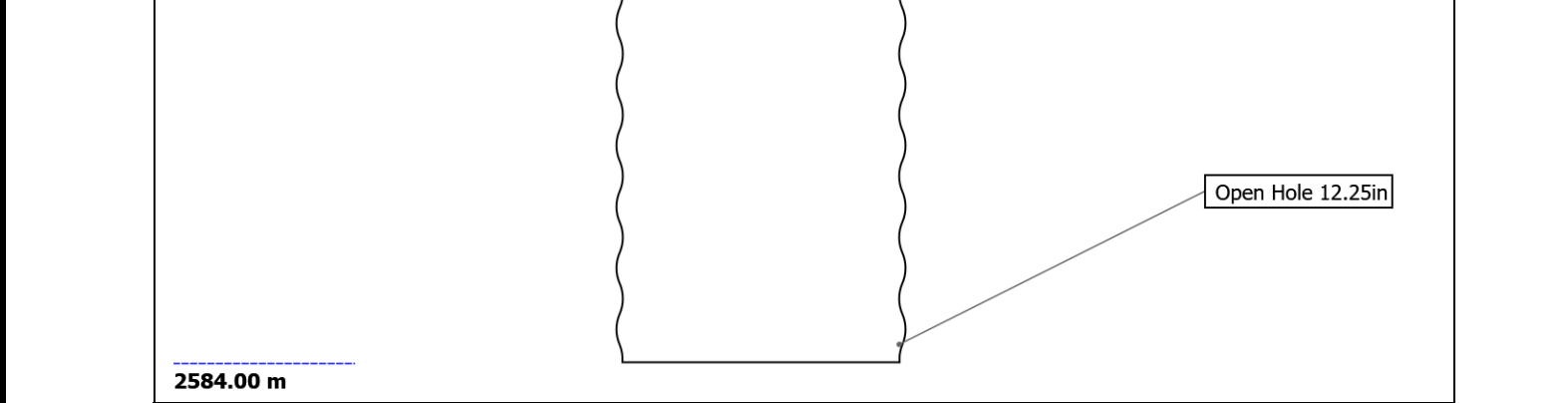
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Well Sketch





Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25		
Top Driller (m)	338	441.5	931	1904		
Bottom Driller (m)	441.5	931	1904	2584		
Casing						
Size (in)	30	20	13.625			
Weight (lbm/ft)	307.09	89.37	59.27			
Inner Diameter (in)	28.063	19.164	12.804			
Grade	X56	N80	P110			
Top Driller (m)	331.1	334.1	335.3			
Bottom Driller (m)	406.6	925.1	1896.7			

Operational Run Summary

Parameter (unit)	Run 8					
Date Log Started	23-Dec-2019					
Time Log Started	06:55:46					
Date Log Finished	25-Dec-2019					
Time Log Finished	13:49:45					
Bit Size (in)	12.250					
Bit Start Depth (m)	1904.00					
Bit Stop Depth (m)	2584.00					
Top Log Interval (m)	1900.20					
Bottom Log Interval (m)	2574.80					
Max Hole Deviation (deg)	0.26					
Azimuth of Max Deviation (deg)	22.82					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	G.Boyd					
Witnessed By	J.Eikenes					
Service Order Number	19SCA0085					

Borehole Fluids

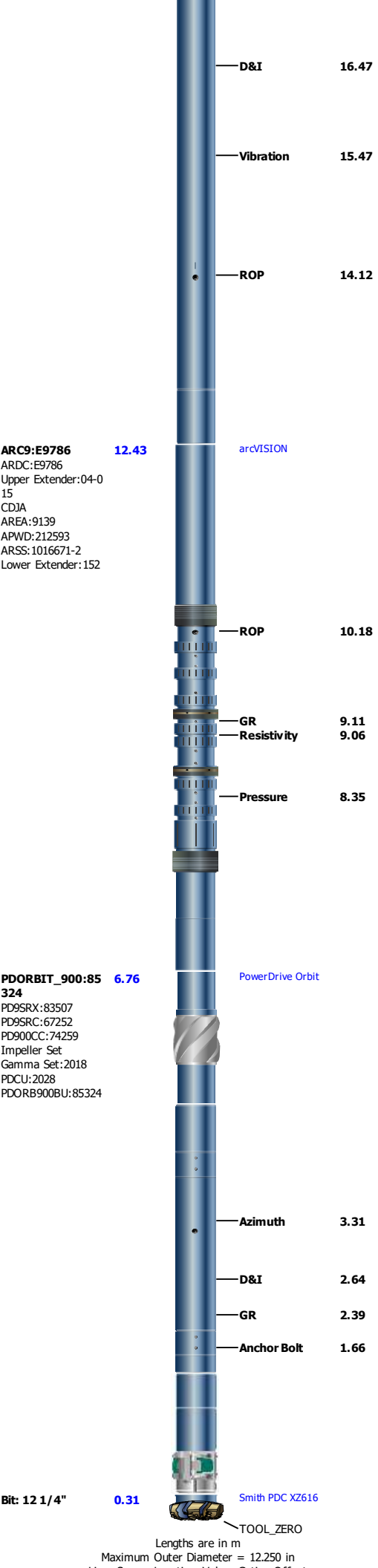
Parameter(unit)	Run 8					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	45					
Source of Sample	Active Tank					
Salinity (ppm)	Zoned					
Density (g/cm3)	Zoned					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.07 @ 17.1					
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 18.06					
RMC @ Meas Temp (ohm.m@degC)	0.1 @ 20					
RM @ BHT (ohm.m@degC)	0.04 @ 45					
RMF @ BHT (ohm.m@degC)	0.04 @ 45					
RMC @ BHT (ohm.m@degC)	0.06 @ 45					
Total Solid (%)						
High Gravity Solids (%)						

Zoned Borehole Fluids

Run 8		
Parameter	Value	Start
Salinity	130260.8	23-Dec-19 6:55:46 AM
Salinity	130260.8	24-Dec-19 3:30:00 AM
Density	1.25	23-Dec-19 6:55:46 AM
Density	1.26	24-Dec-19 3:19:00 PM

Remarks and Equipment Summary

Run 8: Toolstring		Run 8: Remarks
Equip name Length NM Stab: 9 1/2":M 24.15 WS 3732  X/O: 9 1/2":XOS- 21.65 947 TELE900:E9186-I 20.82 W MSSU900:74066 Upper Extender:02-0 6-19 MDC900:E9186-IW MMA:7002 MDI:3770 PMVC:43 PMEA:3792 MTA:3757 MTK900:S-298895-7 775 MSSD900:888081-1 Lower Extender, Lon g:10708-9		All depths referenced to driller's depth. See EOWR for depth tracking details.
		All data from tool memory. All data acquired while drilling
		Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
		Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
		Run 8 and 12 1/4 in. section TD at 2584 m.



Run 8

VISION Resistivity - Dual Frequency

Software Version

Acquisition System		Version	
Maxwell 2019.2		9.2.113335.3100	
Application Patch		DnM_NPD-KAI-2019.2_9.2.114235	
Computation	Description		Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family		9.2.113335.3100
ARC9GammaRayComputat ion	ARC9 Gamma Ray Computation Package for both Real-time and Recorded Mode		9.2.113335.3100
Tool Interface	System Version	Loaded Version	
HSPM	2019.0c.001	9.2.113335.3100	
Tool Elements	Description	Software Version	Firmware Version
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 9.00 Inch Tool Drilling Collar	9.2.113335.3100	V9.7B

Pass Summary

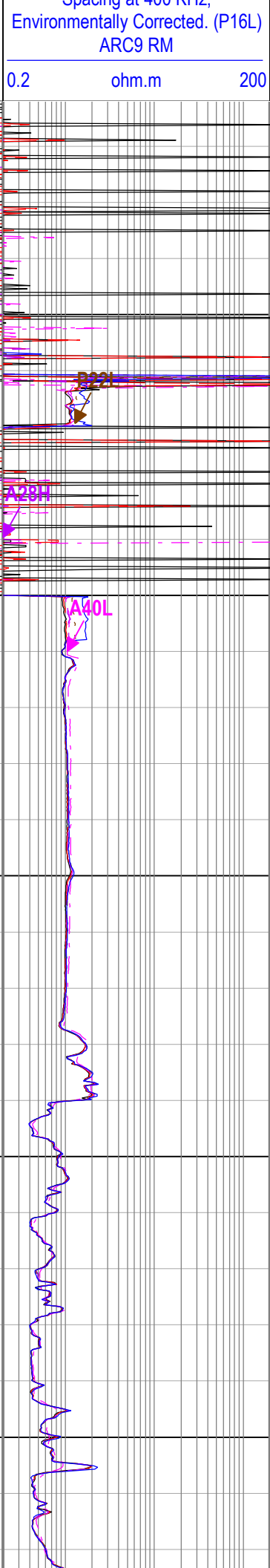
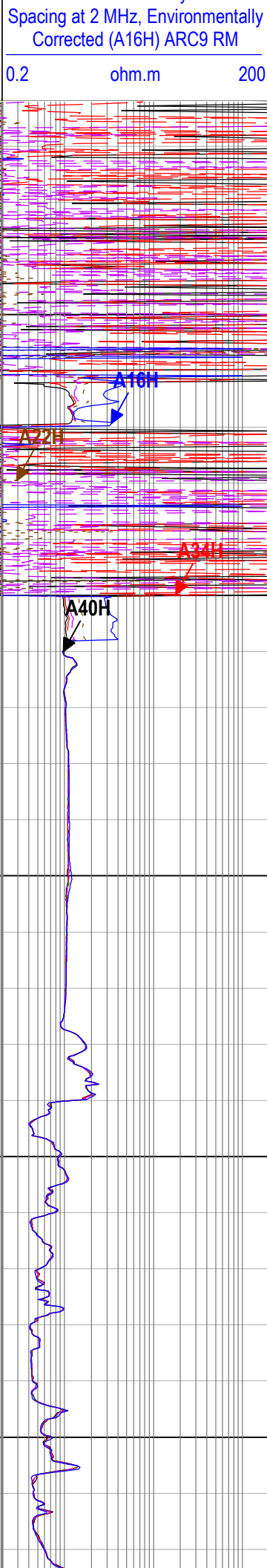
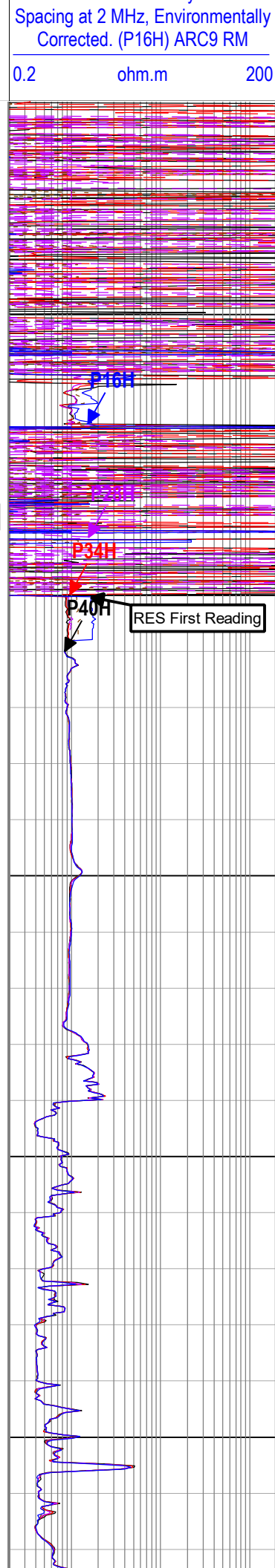
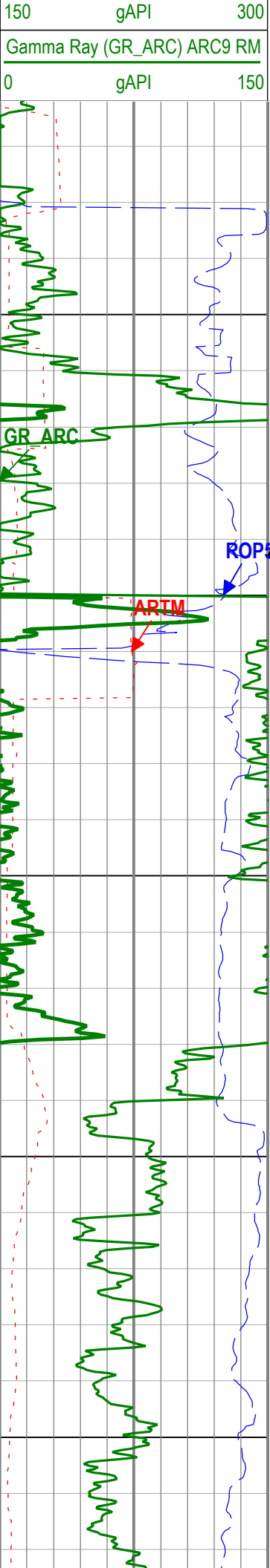
Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 8	Drilling	Down	11.43 m	2583.94 m	23-Dec-2019 6:55:46 AM	25-Dec-2019 1:49:45 PM	Yes
All depths are referenced to toolstring zero							

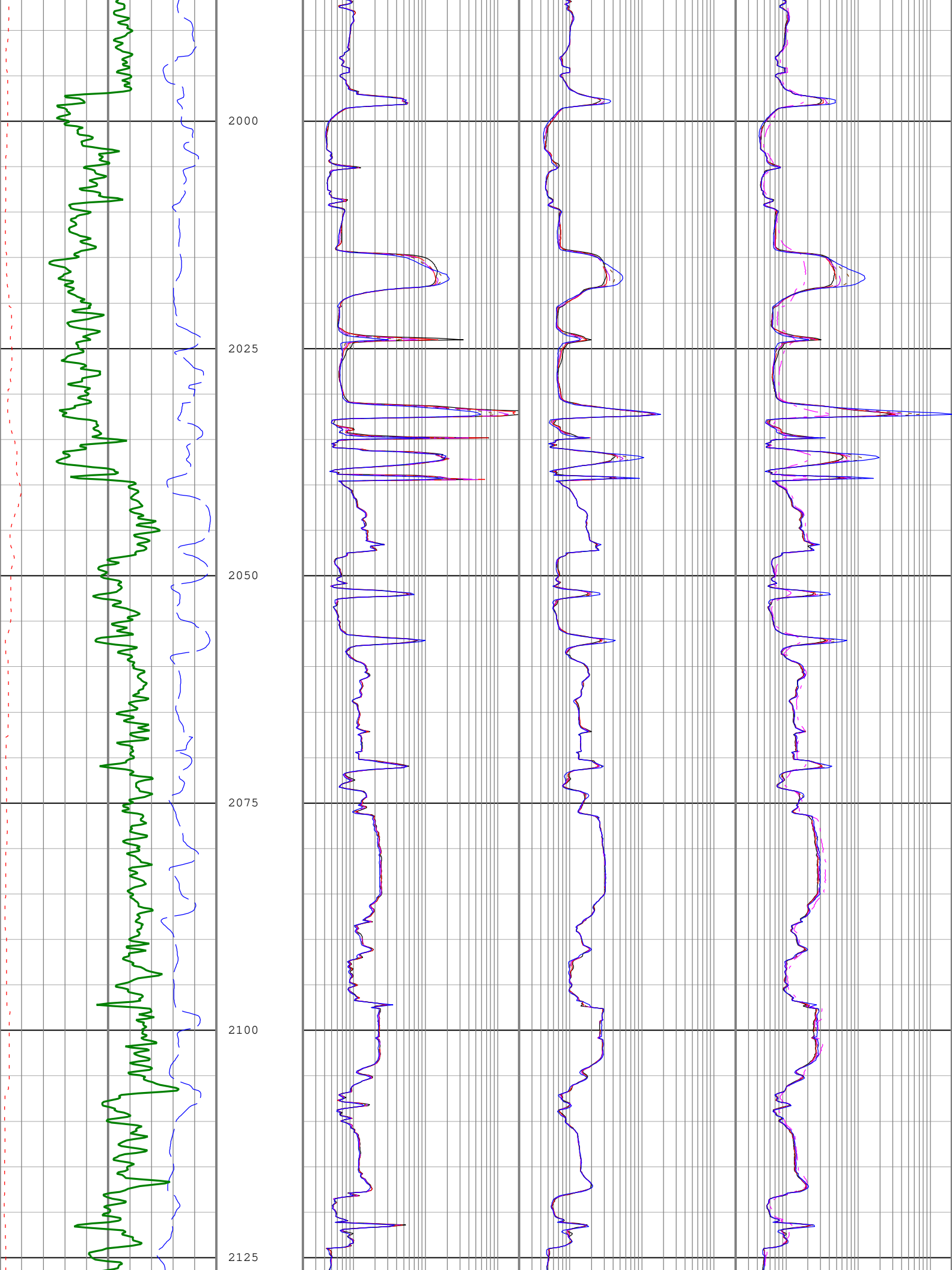
Log

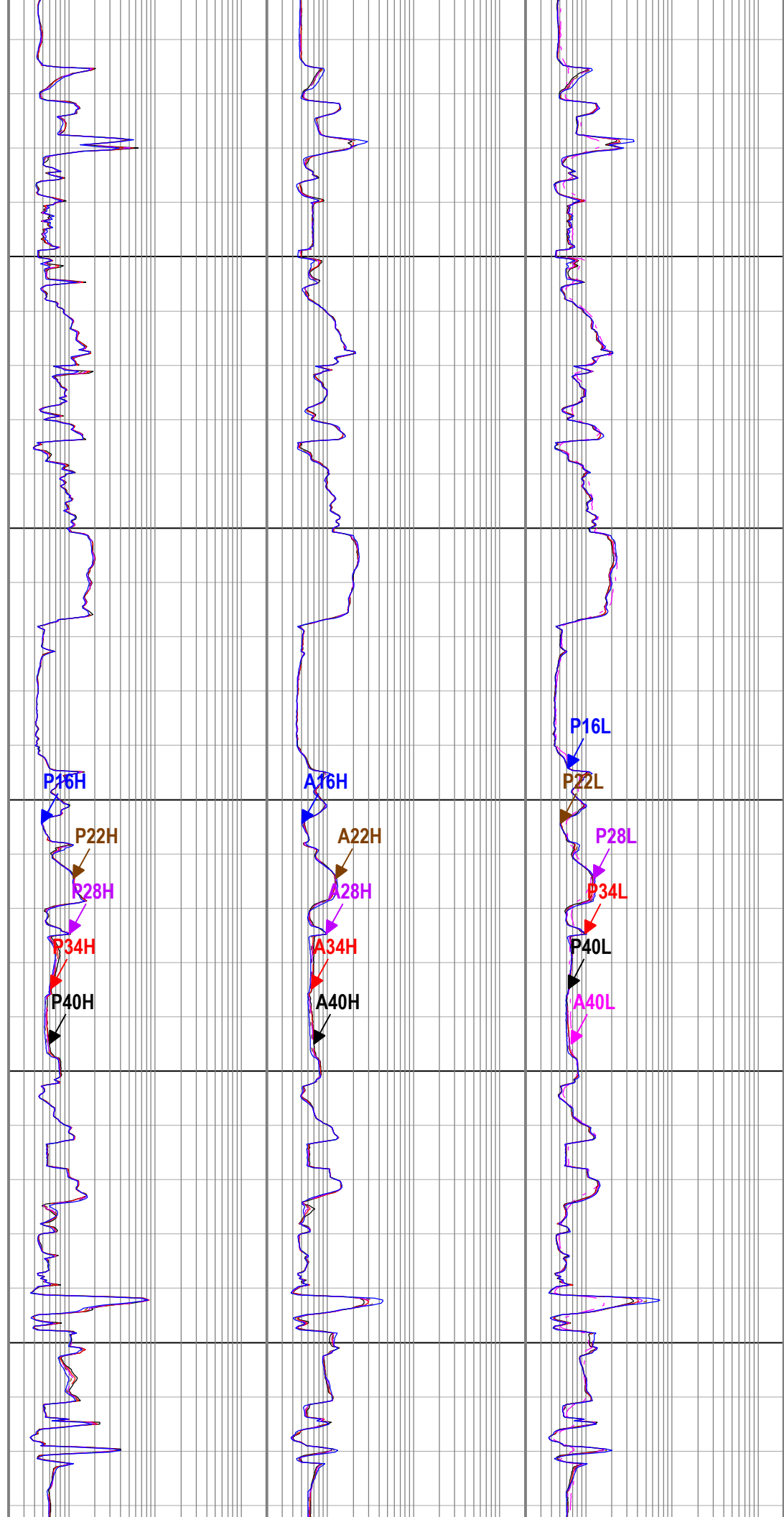
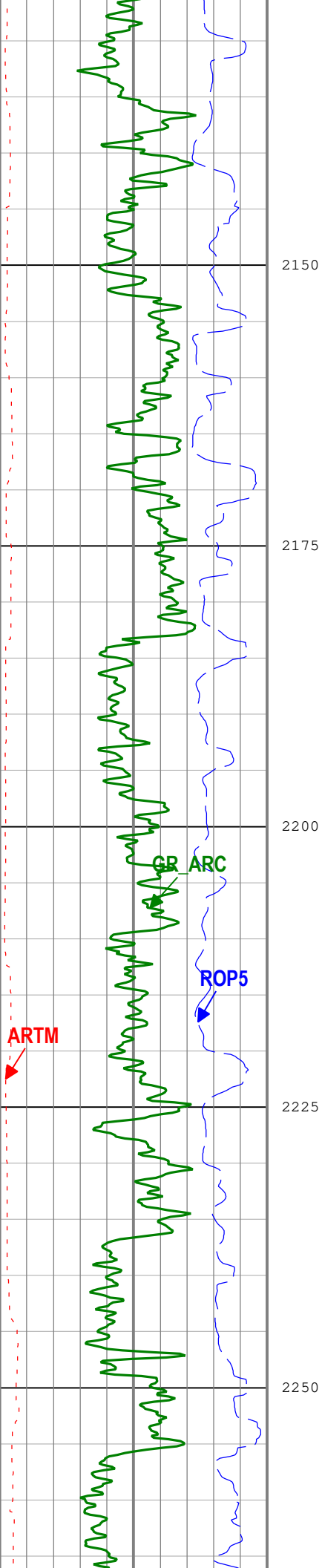
Company:Equinor Well:31/5-7
Run 8: Drilling:S009

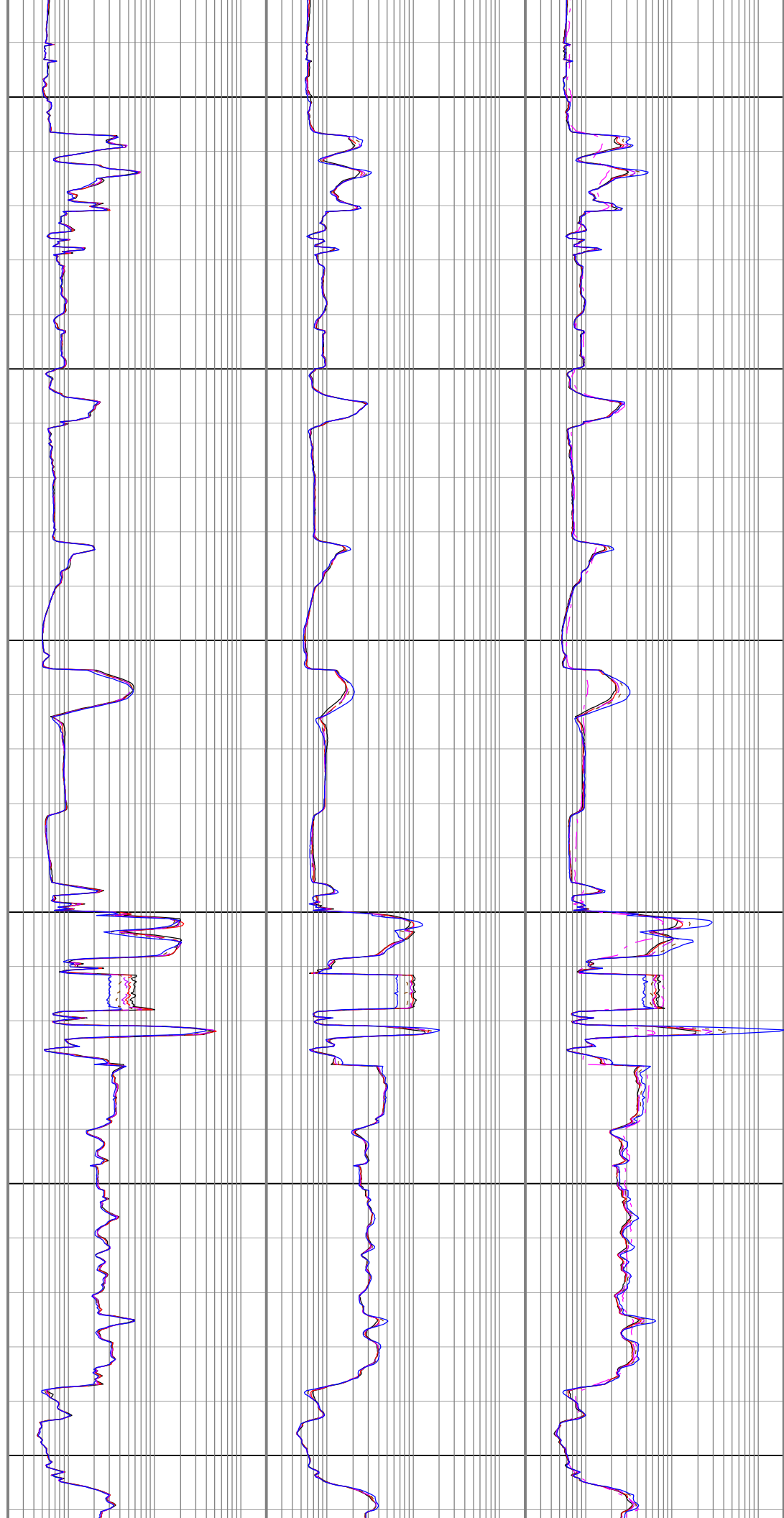
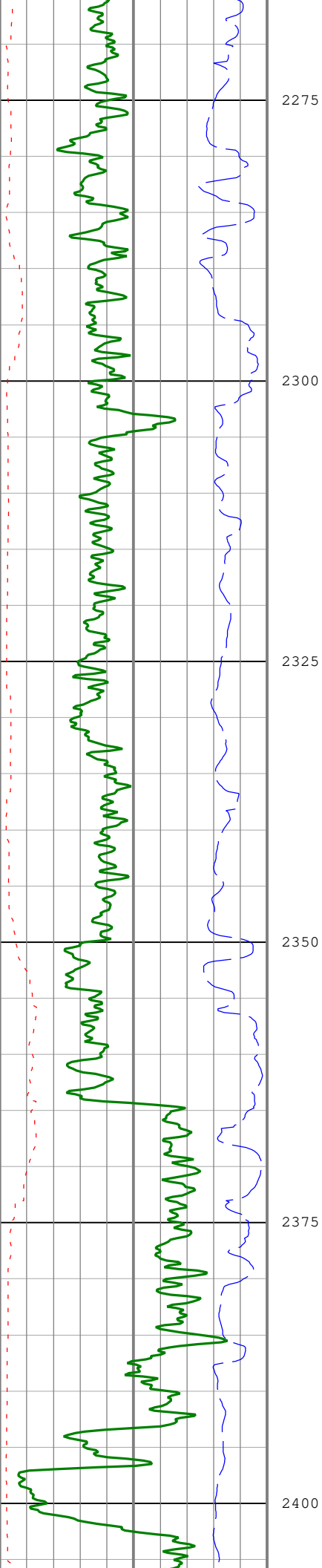
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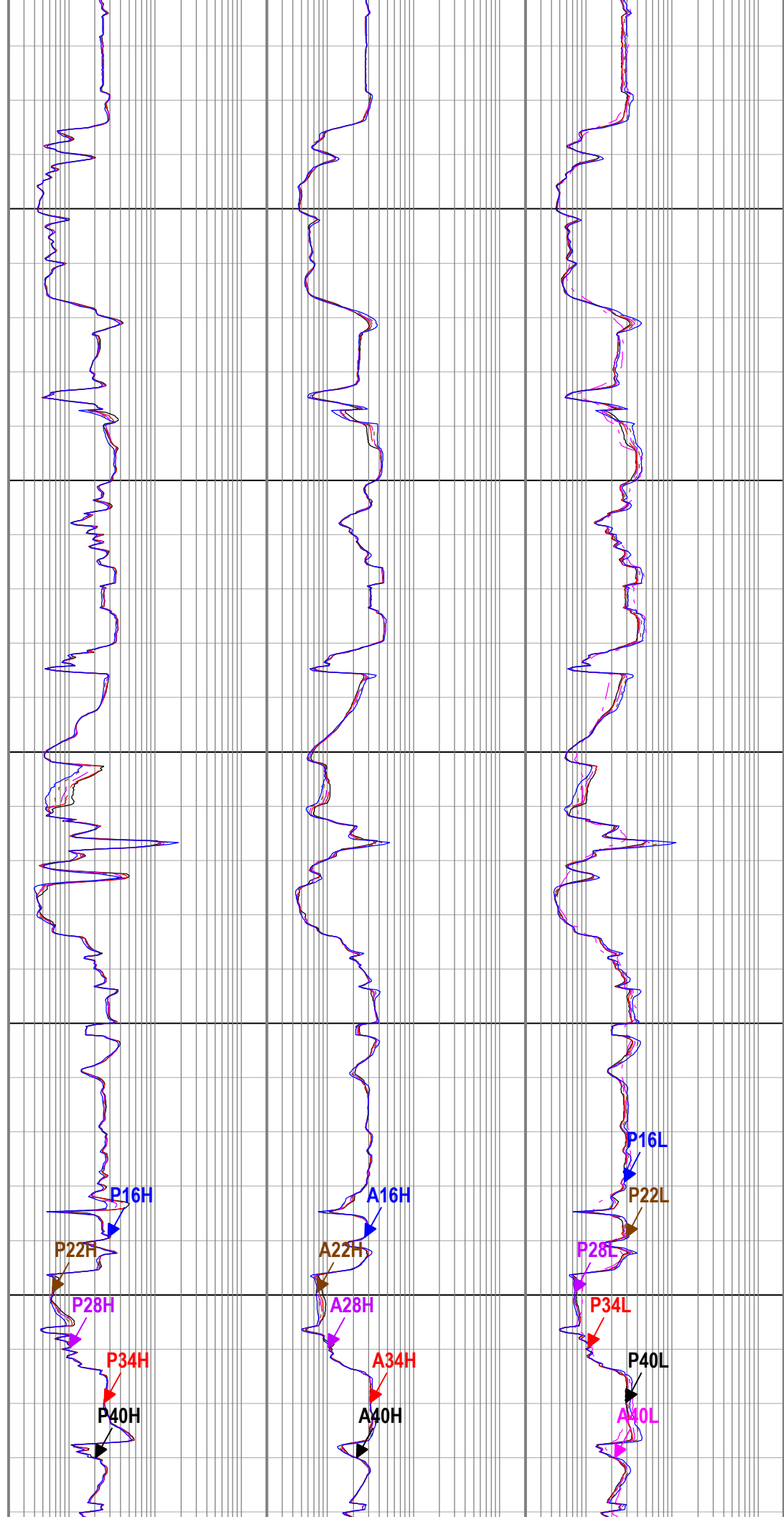
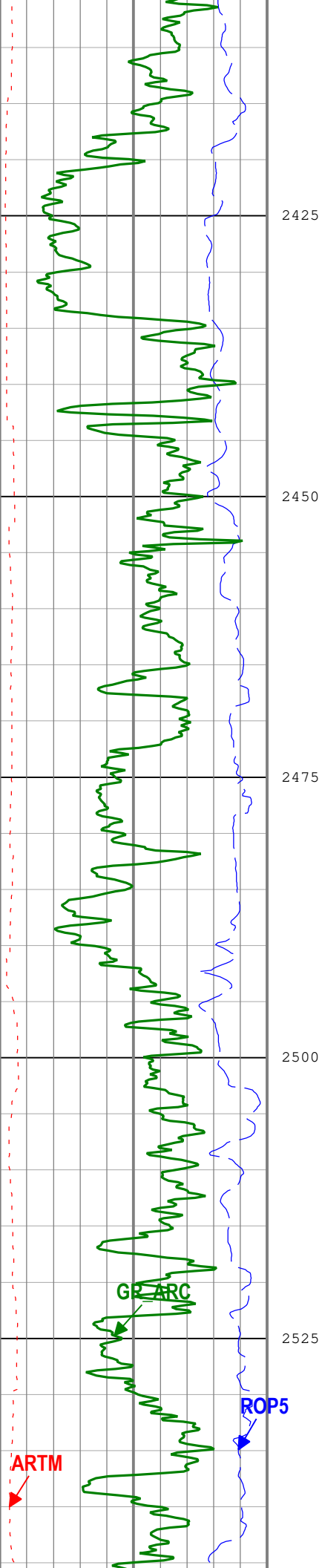
				Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM			
				0.2	ohm.m	200	
				Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM			
				0.2	ohm.m	200	
				Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM			
				0.2	ohm.m	200	
				Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM			
				0.2	ohm.m	200	
				Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM			
				0.2	ohm.m	200	
Resistivity Time After Bit (ARTM) ARC9				Phase Shift Resistivity 16 inch Spacing at 400 KHz			
0 h 10							
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT							
200 m/h 0							
Gamma Ray (GR_ARC) ARC9 RM							

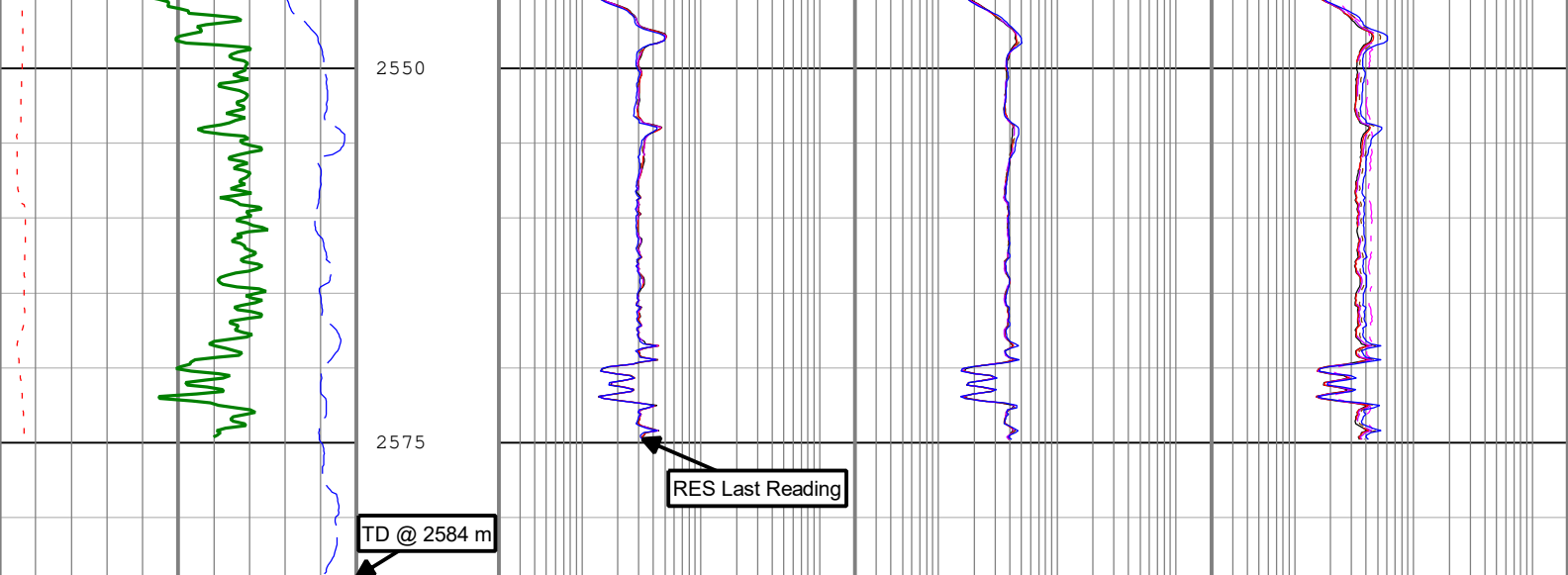












Resistivity Time After Bit (ARTM) ARC9 0 h 10	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM 0.2 ohm.m 200
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT 200 m/h 0	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM 0.2 ohm.m 200
Gamma Ray (GR_ARC) ARC9 RM 150 gAPI 300	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM 0.2 ohm.m 200
Gamma Ray (GR_ARC) ARC9 RM 0 gAPI 150	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM 0.2 ohm.m 200
	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2 ohm.m 200
			Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM 0.2 ohm.m 200

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 30-Dec-2019 10:02:09

Channel Processing Parameters				
Run 8: Parameters				
Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	Time Zoned	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DED	Drilling Fluid Density	Borehole	Time Zoned	g/cm3

	Drilling Fluid Density	Borehole	Time Zone	gamma
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.07	ohm.m

Depth Zone Parameters				
Parameter	Value	Start (m)	Stop (m)	
BS	17.5	1856	1904	
BS	12.25	1904	2584	

All depth are actual.

Time Zone Parameters					
Parameter	Value	Start Time	Stop Time	Start Depth (m)	Stop Depth (m)
BHK	7.12	23-Dec-2019 06:55:46	24-Dec-2019 03:30:00	11.43	2010.59
BHK	6.9	24-Dec-2019 03:30:00	25-Dec-2019 13:49:45	2010.59	2583.94
DFD	1.25	23-Dec-2019 06:55:46	24-Dec-2019 15:19:00	11.43	2300
DFD	1.26	24-Dec-2019 15:19:00	25-Dec-2019 13:49:45	2300	2583.94

All depth are at tool zero.

Tool Control Parameters		
Calibration Report		

ARC9 (Array Resistivity Compensated 900) Calibration - Run 8		
Primary Equipment :		
High Pressure without AIM Receiver	AREA	9139

RESAIRCAL - Resistivity: Air

Master (Time Frame File):		14:29:45 27-Nov-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.526	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.012	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.305	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	3.995	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.920	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	1.422	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-1.320	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	1.370	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-1.331	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	1.348	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.514	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	6.029	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.288	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	4.009	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.909	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.006	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.110	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.046	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.113	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.024	4.100	

GRGAIN - Gamma Ray: Blanket							
Master (Time Frame File):		06:57:38 15-Nov-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200	

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway



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